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Experimental and Computational Investigations of Flow and Mixing in a Single-Annular Combustor Configuration

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Abstract A complementary experimental and computational study of the flow and mixing in a single annular gas turbine combustor has been carried out. The object of the investigation is a generic mixing chamber model, representing an unfolded segment of a simplified Rich-Quick-Lean (RQL) combustion chamber operating under isothermal, non-reacting conditions at ambient pressure. Two configurations without and with secondary air injection were considered. To provide an appropriate reference database several planar optical measurement techniques (time-resolved flow visualisation, PIV, QLS) were used. The PIV measurements have been performed providing profiles of all velocity and Reynolds-stress components at selected locations within the combustor. Application of a two-layer hybrid LES/RANS (HLR) method coupling a near-wall $k-\varepsilon$ RANS model with conventional LES in the core flow was the focus of the computational work. In addition to the direct comparison with the experimental results, the HLR performance is comparatively assessed with the results obtained by using conventional LES using the same (coarser) grid as HLR and two eddy-viscosity-based RANS models. The HLR model reproduced all important flow features, in particular with regard to the penetrating behaviour of the secondary air jets, their interaction with the swirled main flow, swirl-induced free recirculation zone evolution and associated precessing-vortex core phenomenon in good agreement with experimental findings.

Keywords Single-annular swirl combustor · PIV measurements · Quantitative Light-Sheet (QLS) method · LES · RANS · Hybrid LES/RANS methods

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1 Introduction

Injection of the secondary air jets into a swirled flow field of a modern gas-turbine combustion chamber plays a key role with respect to the mixing efficiency. The swirl is important for fuel-mixture generation, providing flame-stabilisation through the formation of a recirculation zone. The quality of the resulting mixing process directly influences the stability of the combustion. The ultimate goal is to achieve a maximum possible homogeneous fuel-mixture being directly associated with a homogeneous combustion temperature distribution contributing strongly to minimization of the pollutants emission. In the present work attention will be restricted to the isothermal mixing process.

The large majority of the experimentally investigated combustor configurations, serving for years as benchmarks for computational methods and turbulence model validation, resemble the tubo-annular combustor geometry representing an axisymmetric idealization of a can-type combustor without secondary air injection, e.g. Roback and Johnson [34], Monmont and Greenhalgh [28] and Palm [30]. Complementary computational studies using LES (Large-Eddy Simulation) and RANS (Reynolds-Averaged Navier–Stokes) methods have been reported for instance in the studies of Pierce and Moin [32], Tang et al. [36], Jester-Zürker et al. [17] and Palm et al. [31]. The experimental investigations of the can-combustor-like geometries exhibiting mixing of the swirled main flow and the radially injected air jets were the topic of the works of Koutmos [20], Heitor and Whitelaw [13] and Koutmos and McGuirk [21]. Whereas the work of Koutmos and McGuirk dealt with an isothermal flow, Heitor and Whitelaw investigated a reacting flow configuration providing the results of the temperature and species (O_2, UHC, H_2, CO_2, CO and NO_x) concentration fields in addition to the mean velocity and turbulence field. The water model of a gas turbine combustor investigated experimentally by McGuirk and Palma [27] was a replica of the model combustor used by Koutmos and Heitor and Whitelaw, differing primarily in number and size of dilution holes and swirler size. The corresponding computational studies using the high Reynolds number versions of the standard $k - \varepsilon$ and differential Reynolds stress models (RSM) in the RANS framework were performed by Lin [24], Koutmos and McGuirk [22] and Mueller and Knill [29]. The latter one relates to the Heitor and Whitelaw's reacting combustor configuration. Lin and Leschziner [25] compared their $k - \varepsilon$ and RSM results of an isolated jet injected into a swirling cross-flow with the experimental database of Gan [7] displaying a more realistic flow pattern obtained by the Reynolds-stress model versions applied.

The tubo-annular combustor types belong to the older generation of gas turbines, being no longer in use. A more common geometry for gas turbines is nowadays the ring-shaped combustor, that is a single-annular combustor, being relevant to the presently investigated configuration. In this configuration the fuel-air mixture is introduced into the combustor through a series of circumferentially arranged swirl nozzles. To systematically study the mixing of the stream introduced centrally into the primary zone through the swirl nozzles and the secondary air injected through the rows of holes situated in the upper and lower combustor walls an experimental facility has been built at the Chair of Fluid Mechanics and Aerodynamics of the Technische Universität in Darmstadt within the framework of the Collaborative Research Center 568 *"Flow and Combustion in Future Gas Turbine Combustion*

Chambers", Fig. 1. This geometry yields highly unsteady behaviour of the flow in general, featuring by coherent structures arising from the swirl nozzles, such as a precessing vortex core (PVC), and its interaction with the secondary air streams with respect to their penetrating characteristics (the configuration resembling the multiple jets in a swirling cross-flow), multiple recirculation zones (swirl-induced flow reversal in the core region, flow recirculation between swirl nozzles and ring-shaped corner bubble) and the wall boundary layer subjected to alternating pressure gradient arising from the varying (spatially and temporally) flow blockage caused by the free back-flow region. The main objective of the present work was the validation of the recently developed hybrid LES/RANS method in this complex flow geometry of practical relevance along the experimental database focussing on the mean velocity and Reynolds stress fields as well as characteristic frequencies pertinent to the PVC phenomenon. The intensity of the mixing process between the main swirled flow and the secondary jets issuing from dilute holes into the combustion chamber was assessed by analysing the secondary air concentration field. Further results displaying correlations between the flow field and the local concentration established by using a POD (Proper Orthogonal Decomposition) analysis are given in Gnirß and Tropea [9]. The present hybrid scheme represents a two-layer model formulation, coupling a RANS model in the wall layer with an LES in the core flow. Hereby, special attention was paid to the treatment of the interface separating the RANS and LES sub-regions concerning its position (determination and control) and the exchange of the RANS- and LES-related variables across it. Both algorithms are formulated in a way providing large range of applicability. Both the RANS model and the Subgrid-Scale (SGS) model in the LES framework utilize the turbulent viscosity as the model variable. Presently a two-equation $k - \varepsilon$ RANS model and the SGS model due to Smagorinsky were adopted, although the method could be easily extended to other eddy-viscosity-based RANS models as well as SGS models, see e.g. Jakirlić et al. [16] and Kniesner [19].

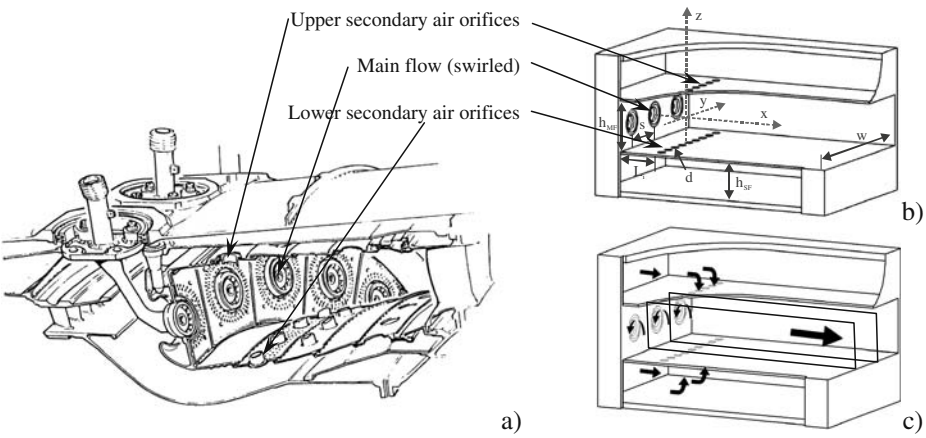

Fig. 1 Example of a single-annular combustor (**a**) and its appropriate simplification pertinent to the present experimental (**b**) and computational investigations (**c**)

Table 1 Geometry parameters of combustor model

Width	$w = 300$ mm
Main channel height	$h_{MF} = 100$ mm
Secondary channel height	$h_{SF} = 50$ mm
Nozzles separation	$s = 100$ mm
Nozzle's outer diameter	$D_{nozzle} = 25$ mm
Injection plane	$L_i = 70$ mm
Orifice diameter	$d = 12$ mm

2 Experimental Facility

The combustor model (Fig. 1; the geometry parameters are summarized in Table 1) was integrated in a test rig consisting of three parallel rectangular channels (main flow channel and two secondary flow channels), which were all driven by one 22 kW blower. The constant global mass flux was reduced using a blow-off pipe and adjusted to the different channels by air regulators. The central main flow (MF) was operated at a mass flow of $\dot{m} = 20$ g/s per nozzle ($D_{nozzle} = 25$ mm) that corresponds to the typical relative pressure loss of $\Delta p/p = 3\%$. The air ($\rho = 1.205$ kg/m^3, $\mu = 1.82 \times 10^{-5}$ N s/m^2) enters the test section through the three swirl nozzles (the geometry with five swirl nozzles was investigated in the work of Gnirß and Tropea [9]) and discharges into the primary zone ($x < 0$) at an ambient pressure of $p_a = 1$ bar. The geometry of the swirl nozzles was taken from an original swirl burner (MTU Aero Engines) scaled up by the factor 2 (Fig. 2a). The coaxial swirl which was generated by tangential rectangular ducts has an overall swirl number of $S \approx 1$. The flow in a tangential duct has the Reynolds number based on the duct height of approximately

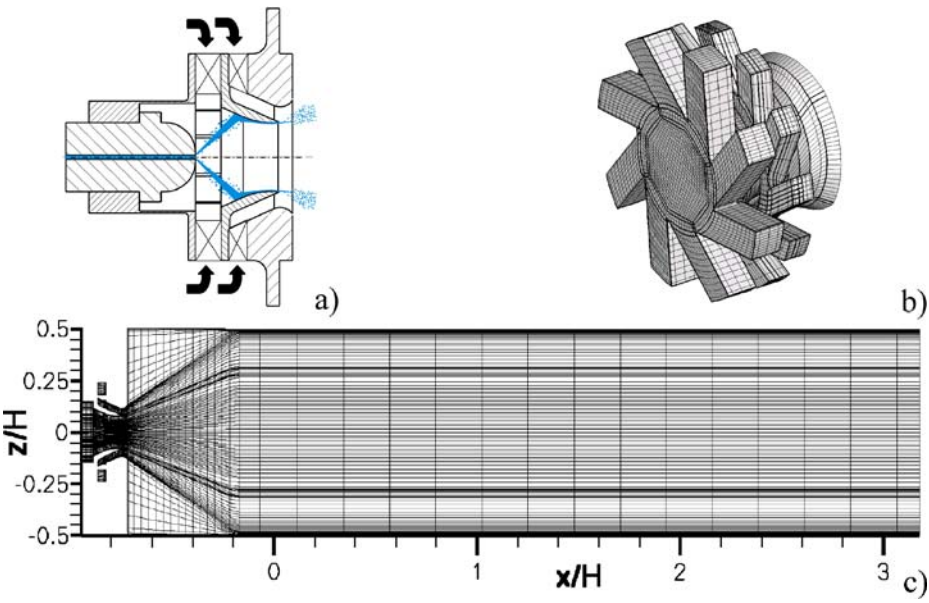

Fig. 2 Schematic (**a**) and corresponding numerical grid (**b**) of the air blast nozzle and numerical grid of the combustor channel (**c**). H is the main channel height ($= h_{MF}$; see Table 1)

11,700. The fuel injection nozzle was deactivated. The Reynolds number of the flow discharging from the outer ring of the swirl nozzle (based on $\overline{v}_{nozzle} = 33.83$ m/s and $D_{nozzle} = 25$ mm) corresponds approximately to 56,000. The Reynolds number of the flow in the main channel is $Re_{MF} = \rho \overline{v}_{MF} h_{MF} / \mu \approx 11,000$ with $\overline{v}_{MF} = 1.66$ m/s. The secondary flow (SF) enters the primary zone from the top and bottom through two rows of orifices. The momentum flux ratio of the secondary air to that of the (swirled) main flow $\left(J = \nu_{SF}^2 / \nu_{MF}^2\right)$ was kept constant: $J = 100$ and 400 (only the first one was investigated computationally). The configuration without secondary air injection ($J = 0$) was also considered.

Measurement techniques Non-intrusive optical measurement techniques have been used based on the measurement of the light scattered from tracer particles. A standard fog generator (SAFEX Fog 2005 Loop) was used with a corresponding fluid to generate these liquid droplets. The resulting aerosol turned out to be almost mono-dispersed with an average particle diameter of $d_p \approx 1.1$ µm, ensuring that the tracer particles could follow the flow up to a limiting frequency of $f_{\max} = 7$ kHz [2]. As shown in Böhm et al. [3] this frequency was sufficient for Laser-Doppler measurements (LDA) within the entire primary zone of the combustor model. However, Gnirß et al. [8] presented planar time-resolved flow-visualizations using a static light-sheet, which was observed by a high-speed camera at 8,000 fps. They pointed out that at the nozzle exit particle-free patches appear and concluded that these patches arose from a limited particle response within the swirl nozzles. It was also shown that this lack of particles in the vortex core could be used to visualize coherent flow structures like the precessing vortex core (PVC) and determine the characteristic frequencies. Böhm et al. [4] suggested to combine this technique with the method of the proper orthogonal decomposition (POD), which is capable of revealing the coherent structures from a set of images. Figure 3 shows the result of such an analysis using the time-resolved images of the case with $J = 0$. The blue (dark) patches in Mode 1 and Mode 2 represent the helical structures of the PVC which is centrally intersected by the light-sheet. A dominant frequency in the spectrum of these two modes, which was calculated from the Fourier transform of

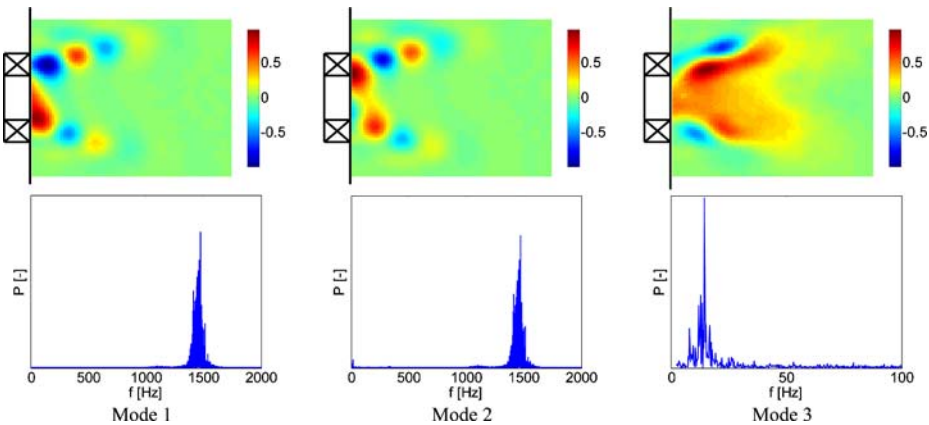

Fig. 3 Visualization of the PVC and corresponding frequencies using POD analysis

the amplitudes $a_i(t)$, appeared at $f_{PVC} \approx 1{,}450$ Hz. Another characteristic behaviour was found in Mode 3 which represents a pulsation of the conical outflow at a 100 times lower frequency $f_{Puls} \approx 14$ Hz. Böhm et al. [4] showed with a spectral analysis of their LDA data that this effect must appear due to an interaction of adjacent swirl nozzles, as it could not be measured after turning off two of the three swirl nozzles. An analysis of the jet penetration behaviour with $J = 100$ reveals that this pulsation also influences the trajectories of the secondary air. They assumed that there were connections between the mixing quality and the underlying flow field, which should be investigated.

PIV measurements were carried out in several horizontal and vertical planes to gain global information of the velocity and Reynolds stress fields. The PIV image processing algorithm was set to an interrogation area size of 32×32 pixels with a window overlap of 50% using window shifting and window deformation. A validation of the correlation peak (peak validation) with an acceptance factor of 1.2 as well as a validation of the velocity range was used to separate outliers. This PIV system was also used for concentration measurements based on Mie-Scattering (Quantitative Light-Sheet—QLS, [38]). For this purpose the setup of Findeisen et al. [6] was slightly modified for measurements to be conducted in the longitudinal measurement situation. The readers interested in the details about data procession are referred to Gnirß and Tropea [9].

The velocity vector plots obtained by the PIV method in the horizontal $x - y$ ($z = 0$) plane within the mixing chamber indicate a non-periodic flow behaviour between adjacent swirl nozzles, Fig. 6. These results, representing the average of 200 pictures, were obtained by using the test rig with five swirl nozzles. Such a non-periodicity, characterized by a highly unsteady flow reversal between adjacent swirl nozzles, intensifies with increase of the momentum flux ratio J. It is especially visible in the case with $J = 400$ (not treated computationally in the present work). In the work of Gnirß [10] this phenomenon was traced back to the presence of the side walls (positioned at $y = \pm 250$ mm from the central swirl nozzle), still affecting significantly the flow issuing from the centrally ($y = 0$) located swirl nozzle. In the simulations only the central swirl nozzle was accounted for with the periodic boundary conditions applied at the planes positioned symmetrically between two neighbouring nozzles, Fig. 1c. The flow behaviour described can certainly not be captured computationally in such a way. Consequently, some departures of the computational results from the experimental ones can be explained to a certain extent by using these arguments.

3 Computational Model

The continuity ($\partial \overline{U}_i / \partial x_i = 0$), momentum and concentration equations governing incompressible flow and mixing under the constant fluid property conditions read:

$$\frac{\partial \overline{U}_i}{\partial t} + \frac{\partial \left(\overline{U}_i \, \overline{U}_j \right)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \overline{P}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\tau_{ij}^{v} + \tau_{ij}^{t} \right) \tag{1}$$

$$\frac{\partial \overline{C}}{\partial t} + \frac{\partial \left(\overline{U}_j \, \overline{C} \right)}{\partial x_j} = \frac{\partial}{\partial x_j} \left(d_j^{v} + d_j^{t} \right) \tag{2}$$

Here, $\tau_{ij}^{v} = 2\nu\overline{S}_{ij}$ (with $\overline{S}_{ij} = 0.5\left(\partial\overline{U}_i/\partial x_j + \partial\overline{U}_j/\partial x_i\right)$ being the rate of strain tensor) and $d_j^{v} = \left(\nu/Sc\right)\partial\overline{C}/\partial x_j$, with $Sc = 0.71$, represent the viscous stress tensor and viscous concentration flux, whereas the turbulent stress tensor τ_{ij}^{t} and turbulent concentration flux d_j^{t} are to be modelled. ρ and ν are fluid density and kinematic viscosity respectively. The rationale and the most important features of the turbulence model used in the present work are outlined in the following subsections.

Hybrid LES/RANS (HLR) model In the present hybrid LES/RANS formulation, the RANS model covers the near-wall region and the LES model the remainder of the flow domain. The unsteadiness in the RANS sub-region is largely caused by the intensive interaction with the unsteady LES flow field. Both methods share the same temporal resolution. The equations governing the velocity (Eq. 1) and mixture fraction (Eq. 2) fields operate as the Reynolds-averaged Navier–Stokes equations in the near-wall layer ($\overline{U}_i$ and $\overline{C}$ represent the ensemble-averaged fields) or as the filtered Navier–Stokes equations in the outer layer ($\overline{U}_i$ and $\overline{C}$ represent the spatially filtered fields). The turbulent stress tensor τ_{ij}^{t} in Eq. 1 representing either the subgrid-stress tensor $\left(\tau_{ij}^{t} \equiv -\tau_{ij} = 2\nu_t\overline{S}_{ij} - \tau_{kk}\delta_{ij}/3\right)$ or the Reynolds-stress tensor $\left(\tau_{ij}^{t} \equiv -\overline{u_i u_j} = 2\nu_t S_{ij} - 2\rho k\delta_{ij}/3\right)$ is expressed in terms of the mean strain tensor via Boussinesq's relationship. The turbulent concentration flux in the equation governing the mixture fraction field is modelled using the simple gradient diffusion hypothesis $d_j^{t} = \left(\nu_t/Sc_t\right)\partial\overline{C}/\partial x_j$, with $Sc_t = 0.7$. The equations governing the velocity and concentration fields in the hybrid LES/RANS framework read:

$$\frac{\mathrm{D}\overline{U}_i}{\mathrm{D}t} = -\frac{1}{\rho}\frac{\partial\overline{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial\overline{U}_i}{\partial x_j} + \frac{\partial\overline{U}_j}{\partial x_i}\right)\right] \tag{3}$$

$$\frac{\mathrm{D}\overline{C}}{\mathrm{D}t} = \frac{\partial}{\partial x_j}\left[\left(\frac{\nu}{Sc} + \frac{\nu_t}{Sc_t}\right)\frac{\partial\overline{C}}{\partial x_j}\right] \tag{4}$$

In the present two-layer hybrid LES/RANS scheme the coupling of the instantaneous LES field and the ensemble-averaged RANS field at the interface is realized via the turbulent viscosity, which makes it possible to obtain solutions using one system of equations. This means practically that the governing equations (1) and (2) are solved in the entire solution domain irrespective of the flow sub-region (LES or RANS). Depending on the flow zone, the hybrid model implies the determination of the turbulent viscosity ν_t either from a $k-\varepsilon$ RANS model: $\nu_t = C_\mu f_\mu k^2/\varepsilon$ or from the subgrid-scale (SGS) model in the LES formulation: $\nu_t = \nu_{SGS} = (C_S\Delta)^2\left|\overline{S}\right|$. The Smagorinsky constant C_s takes the value of 0.1. $\Delta = (\Delta x \times \Delta y \times \Delta z)^{1/3}$ represents the filter width and $\left|\overline{S}\right| = \left(\overline{S}_{ij}\overline{S}_{ij}\right)^{1/2}$ the strain rate modulus. The near-wall variation of the turbulent viscosity ν_t is obtained from a $k-\varepsilon$ RANS model implying solving of the following two transport equations:

$$\frac{\partial k}{\partial t} + U_j\frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + P_k - \varepsilon \tag{5}$$

$$\frac{\partial\,\varepsilon}{\partial\,t}+U_j\frac{\partial\,\varepsilon}{\partial\,x_j}=\frac{\partial}{\partial x_j}\left[\left(\nu+\frac{\nu_t}{\sigma_\varepsilon}\right)\frac{\partial\,\varepsilon}{\partial\,x_j}\right]+C_{\varepsilon,1}P_k\frac{\varepsilon}{k}-f_\varepsilon\,C_{\varepsilon,2}\frac{\varepsilon^2}{k}-P_{\varepsilon,3} \tag{6}$$

The near-wall and viscous damping functions (f_μ, f_ε) and the production term $P_{\varepsilon,3}$ (Eqs. 5 and 6), are presently modelled in line with the proposal of Launder and Sharma [23] with ε representing the isotropic part of the total viscous dissipation rate $\left(\varepsilon\rightarrow\tilde{\varepsilon}=\varepsilon-2\nu\left(\partial k^{1/2}/\partial x_k\right)\left(\partial k^{1/2}/\partial x_k\right)\right)$ taking zero value at the wall ($\tilde{\varepsilon}_{wall}=0$). This near-wall model is a straightforward extension of the standard high-Reynolds number $k-\varepsilon$ model (used presently for the RANS and hybrid LES/RANS computations of the second combustor case, Sections 4) aiming at capturing of the enhanced viscosity influence onto turbulence in the immediate wall vicinity (viscous sub-layer and buffer layer). The differences (instead of the total viscous dissipation rate ε, its "isotropic part" $\tilde{\varepsilon}$ is used as the scale-supplying variable; addition of a model production term originating from the so-called gradient production term $P_{\varepsilon,3}$ into the dissipation equation; introduction of a damping function f_μ into the eddy-viscosity model formulation) are limited only to the near-wall region. Solving the equations for k and ε in the RANS sub-domain (being necessary for the turbulent viscosity determination, Eqs. 3 and 4) requires appropriate boundary conditions at the RANS–LES interface. The approach used presently is based on the equality of the RANS interface-values for k and ε with corresponding SGS values. This condition, originating from the equivalence of the total stresses at both sides of interface (the RANS total stress, similar to the LES one, comprises both resolved and modelled parts) - $\tau_{tot}^{LES}=\tau_{tot}^{RANS}$, reduces to the equality of their modelled contributions by assuming the continuity of the resolved fractions across the interface (*ifce*), Hanjalić [12] and Temmerman et al. [37]: $-2\nu_{SGS}\overline{S}_{ij}=-2\nu_t S_{ij}$. Finally, the condition applied at the interface implies the equality of the (modelled) turbulent viscosities:

$$\nu_{t,ifce}|_{RANS-side}=\nu_{SGS,ifce}|_{LES-side} \tag{7}$$

The statement concerning the equivalence of the resolved fractions in the immediate interface vicinity originates from the fact that the HLR method deals with a unique, LES-like velocity field in the entire solution domain independent of the sub-region. It should also be noted that this is not entirely true, as we have also experienced in the process of the method validation, but the differences are very small. Introduction of the resolved contributions into the matching criteria led to discontinuities (kinks) at the profiles of turbulent quantities across the interface. These are the consequence of the strong damping of LES unsteadiness in the RANS sub-region and its weaker receptivity to the turbulence fluctuations. Furthermore, accounting for the resolved parts would require their evaluation at each time step, the procedure being not in accordance with requirements for an efficient method.

Because k and ε are not provided within the LES sub-domain in the case of the subgrid-scale model of Smagorinsky, their SGS values are presently estimated utilizing the local equilibrium hypothesis assuming the equality between the production, dissipation and energy flux through the cutoff and the analogy to the eddy-viscosity-based RANS modeling in line with the proposal of Mason and Callen [26]:

$$k_{SGS}=\frac{(C_S\Delta)^2\left|\overline{S}\right|^2}{\sqrt{C_\mu}}\,,\quad \varepsilon_{SGS}=(C_S\Delta)^2\left|\overline{S}\right|^3 \tag{8}$$

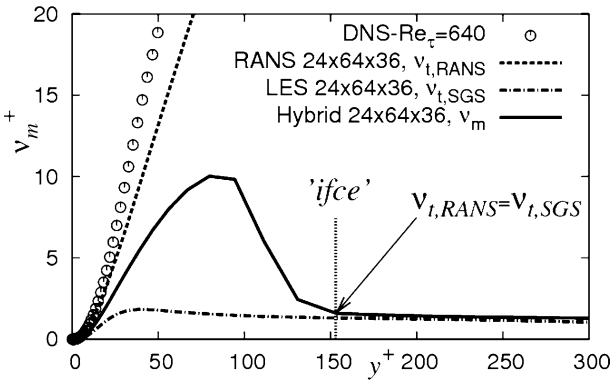

Fig. 4 Variation of modelled turbulent viscosity $(\nu_m^+ = \nu_t/\nu)$ across the interface (*ifce*) in a fully-developed channel flow (DNS: [1])

with $C_\mu = 0.09$. The algorithm being used to numerically ensure the fulfilment of the condition expressed by Eq. 7, resembling the well-known procedure for setting the value of a variable at a computational node, is explained as follows. The RANS equations for k and ε are solved in the entire flow field, but with the discretization coefficients taking zero values in the LES sub-region ($A_{k,SGS} = 0$). By manipulating appropriately the source terms in the corresponding discretization equations $\left(\sum_k A_k + S_P\right)_{SGS} \Phi_{P,SGS} = \sum_k A_k \Phi_{k,SGS} + S_{U,SGS}$ by taking $S_{U,SGS} = 10^{30}\Phi_{SGS}$ and $S_{P,SGS} = 10^{30}$, with $\Phi_{SGS} \equiv k_{SGS},\ \varepsilon_{SGS}$, their numerical solution in the framework of the finite volume method provides the interface values of the k_{RANS} and ε_{RANS} being equal to the corresponding SGS values. By doing so, the boundary condition at the LES/RANS interface (*ifce*), expressed by Eq. 7, is implicitly imposed without introduction of any empirical "transitional" function. Both the RANS model and the SGS model are used in their original form without any further adjustment or modification. This is illustrated in Fig. 4 displaying the variation of the modelled turbulent viscosity across the interface in a fully-developed channel flow. One can clearly see the damping of the RANS viscosity towards the one of the SGS motion. A gradual adjustment of the turbulent viscosities at both interface sides to each other (within the numerical method applied) is ensured, providing in such a way their smooth transition.

One of the advantages of a zonal approach is the possibility to predefine the LES–RANS interface. However, in unknown flow configurations, this could be a difficult issue. Therefore, a certain criteria expressed in terms of a control parameter should be introduced. Presently, the following control parameter

$$k^* = \left\langle \frac{k_{\text{mod}}}{k_{\text{mod}} + k_{res}} \right\rangle \tag{9}$$

is adopted, representing the ratio of the modelled ($k_{\text{mod}} = k_{SGS}$) to the total turbulent kinetic energy in the LES region (with k_{res} representing the resolved part, whose evaluation requires the on-line averaging), averaged over all grid cells bordering the interface on the LES side. Thus, the interface will be positioned in accordance with the requirement for the k^*-parameter, i.e. for the modelled turbulent kinetic energy, to stay under a certain limit which is to be prescribed. A typical value of 20% was adopted in the present work, corresponding approximately to the reference value an LES resolution should comply with, Pope [33]. As soon as the

k^*-value exceeds 20% (implying a coarser grid at this position, being inadequate for LES to be applied), the interface is moved farther from the wall (the RANS sub-region will be enlarged) and in the opposite direction when the value goes below 20% (denoting a finer grid at the respective position being suitable for LES to be appropriately applied). This additionally ensures that in the limit of a very fine grid (low level of the residual turbulence) LES is performed over most of the solution domain. In contrast, in the case of a coarse grid, RANS prevails. As the interface separates the near-wall region from the reminder of the flow, it would be suitable to choose a wall-defined parameter for denoting the interface location. For present computations of the model combustor (Fig. 1b and c), both dimensionless wall distances z^+ (made dimensionless by applying the wall units ν and U_τ) and z/Δ (with $\Delta = (\Delta x \times \Delta y \times \Delta z)^{1/3}$ representing the corresponding mesh size) were adopted (see Section 4 for more details). Despite possible difficulties with respect to the definition of z^+ in flow domains where the wall shear stress $\left(\tau_w = \rho U_\tau^2\right)$ approaches zero, as e.g. in separation and reattachment regions, no problems in the course of the computations have arisen. It should be noted, that the interface z^+ doesn't represent a model parameter in the HLR method. It only denotes, i.e. "labels" the computational nodes at which the prescribed value of k^* is obtained, i.e. "detected". Figure 5a illustrates the behaviour of the interface position in a fully-developed channel flow by displaying its temporal variation in terms of dimensionless wall distance y^+. In this flow being two-dimensional in the mean, the wall-normal coordinate is denoted by y, unlike z in the combustor configuration (Fig. 1b). The final position of the interface obtained at the end of this computation corresponds to $y^+_{ifce} \approx 50$. It should be noted that the grid used here, comprising $(N_x, N_y, N_z) = (48, 64, 72)$ grid cells, is substantially finer than one used for the HLR simulation whose results are shown in Fig. 4. In Fig. 5b the variation of the instantaneous k^* value, determined from Eq. 9, along the interface whose wall distance corresponds to $y^+_{ifce} \approx 50$ is depicted. If the interface position is defined by the non-dimensional wall distance y/h (with h representing the channel height) the final value of the control parameter k^* (Eq. 9) corresponds exactly to the prescribed value of 20%. On the other hand, if the wall distance y^+ was applied for the definition of the interface position a certain variation of the instantaneous k^* value around the averaged, i.e.

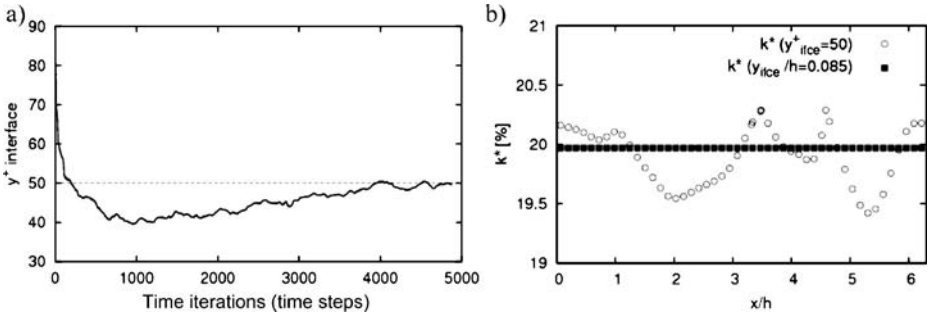

Fig. 5 Temporal variation of the interface position in terms of dimensional wall distance y^+ (**a**) and variation of the instantaneous k^* value around the averaged, i.e. prescribed one (20%) along the interface (**b**) in a fully-developed channel flow

prescribed one was obtained due to changes in wall shear stress value. These changes are fairly weak in the case of a fully-developed channel flow. In the case of the flows affected e.g. by separation these variations are much more intensive. However, the prescribed averaged value of 20% will be always achieved. The determination and control of the interface location is numerically realized in accordance with the following procedure. The computation is started with an initially prescribed interface whose position corresponds to a certain dimensionless wall distance, z^+ or z/Δ. The value of the k^* parameter is evaluated after every (typically) 10–15 time steps (at the very beginning of the iterative procedure this evaluation can be performed after several hundreds temporal iterations) in accordance with Eq. 9 and compared to its prescribed value of 20%. Depending on its departure from this value, the interface is shifted appropriately, as explained above. This procedure is repeated until the prescribed k^* value is reached. After the interface converges towards a position corresponding to $k^* \approx 20\%$ the algorithm can be deactivated.

The present HLR model has been thoroughly validated in configurations of different geometrical complexity featured by flow separation (from sharp-edged and continuous curved surfaces) and strong heating under the conditions of constant and variable fluid properties. Readers interested in more details about the present hybrid LES/RANS method are referred to Jakirlić et al. [16], Kniesner et al. [18] and Kniesner [19].

RANS models The RANS calculations of the case without secondary jets were performed using a robust, eddy-viscosity-based k–ε–ζ–f turbulence model (Hanjalić et al. [14]), denoted by KZF throughout the manuscript. This model represent a near-wall model implying the integration of the governing equations through the entire near wall region, i.e. the application of the exact wall boundary conditions. Similar to the LS model, employed to model the near-wall region in the hybrid LES/RANS framework, the standard k–ε model serves as the high Reynolds number asymptote also in this case. The k–ε–ζ–f is a four-equation turbulence model. Simultaneously with the equations governing k and ε the equations for the variable ζ and elliptic function f are to be solved. The model relies on the Durbin's elliptic relaxation concept providing a continuous modification of the homogeneous pressure-strain process as the wall is approached to satisfy the wall conditions, thus avoiding the need for any wall topography parameter. The variable ζ represents the ratio ν^2/k (ν^2 is a scalar property in the Durbin's ν^2–f model, which reduces to the wall-normal stress in the near-wall region) providing more convenient formulation of the equation for ζ and especially of the wall boundary conditions for the elliptic function f. This model approach represents a further contribution towards more robust use of advanced closure models. The most important difference between this model and the LS model is in the improved modelling of the near-wall region, related primarily to the model formulation of the eddy-viscosity. Unlike the LS model, which accounts only for the viscosity effects in the near-wall region, the k–ε–ζ–f model accounts also for the Reynolds stress anisotropy through better modelling of the characteristic velocity scale ϑ and through introduction of the Kolmogorov scale $\tau_K = (\nu/\varepsilon)^{1/2}$ into the characteristic time scale τ into the model of $\nu_t \propto \vartheta^2\tau$: $\vartheta^2 = k$ and $\tau = k/\varepsilon$ in the LS model and $\vartheta^2 = \zeta k$ and $\tau \propto \max(\tau_K; k/\varepsilon)$ in the k–ε–ζ–f model. The readers interesting in more details are referred to the original publication of Hanjalić et al. [14]. The second combustor configuration accounting for the secondary air injection

has been computed using the basic, high-*Re* k–ε model (denoted by KE throughout the work) in conjunction with the standard wall functions for treating the near-wall region. The three RANS models used in the present work are highly complementary.

Numerical method All computations were performed with the in-house computer code FASTEST [5] based on a finite volume numerical method for solving both the three-dimensional filtered and Reynolds-averaged Navier–Stokes equations on block structured, body fitted, non-orthogonal meshes. Block interfaces are treated in a conservative manner, consistent with the treatment of the inner cell faces. A cell-centered (collocated) variable arrangement and Cartesian vector and tensor components are used. The well-known SIMPLE algorithm is applied for coupling the velocity and pressure fields. The convective terms in all equations are discretized by a second-order central differencing scheme, implemented by utilizing the deferred correction approach. The time discretization is accomplished applying the second order (implicit) Crank–Nicolson method. The adopted time step in simulations of both combustor configurations without and with secondary injection corresponds to $\Delta t = 5 \times 10^{-6}$ s The resulting CFL number takes approximately unity value in the swirl generator. Its value in the mixing chamber amounts to 0.1. The RANS computations were performed in the steady manner (the initially started unsteady computations ended up with the steady solutions).

4 Results and Discussion

Selected results obtained by computing both configurations without ($J = 0$) and with secondary air injection ($J = 100$) are illustrated in Figs. 8, 9, 10, 11, 12, 13 and 14. The computational domain comprises only one air-blast nozzle with side boundaries located in the middle between two consecutive nozzles, Fig. 1c. This enables application of the periodic boundary conditions in the spanwise direction (y-direction in Fig. 1b). The bulk velocity in such a main channel corresponds to $\overline{v}_{MF} = 1.66$ m/s leading to the Reynolds number $Re_{MF} \approx 11{,}000$. The flow in the swirl nozzle, i.e. swirl generator was accounted for (see Fig. 2b for appropriate numerical grid). The nozzle consists of two rings with eight and 12 rectangular channels respectively. At each channel inlet a uniform velocity profile obtained from a given mass flow rate was prescribed. The air was sufficiently perturbed when flowing through the nozzle so that no turbulent fluctuations had to be imposed at the inlet. The generated central and annular nozzle streams discharge into the main channel swirling in the same sense. The outlet plane was located at four main channel heights (≈400 mm) from the nozzles. The convective boundary conditions were applied there.

Combustor without secondary air injection ($J = 0$) The numerical grid (Fig. 2c) consists of 180 blocks comprising in total 460,000 grid cells. Such a relatively coarse grid was intentionally designed keeping in mind that the basic motivation behind the HLR method is to obtain results at a coarser grid whose quality is comparable with that of a well-resolved LES. This grid represents an appropriately coarsened version of the grid containing 3.7 million grid cells Sternel et al. [35] have applied for their LES study. Detailed analysis with respect to an optimum grid size, time

step and convergence criteria has been performed. However, as they computed the case with $\dot{m} = 50\ \mathrm{g/s}$ per nozzle, being 2.5 times higher than the mass flow rate considered here $\dot{m} = 20\ \mathrm{g/s}$, a direct comparison with the present computational results was not possible. It should also be noted that the grid they used would be even finer for the case with lower mass flow rate, i.e. lower flow Reynolds number. We adopted their grid and coarsened it appropriately concerning its use in conjunction with a hybrid LES/RANS method. As the basic idea behind a zonal hybrid method with RANS covering the near-wall region is to obtain results of the quality of a fine LES by using coarser spatial and temporal resolution, the use of HLR in conjunction with an LES-like, fine grid wouldn't make sense. In the present zonal model a discrete, flow-dependent interface changes its position in the course of the computation in accordance with the prescribed value of the control parameter $k^* = 20\%$ (see Section 3 for more details). In the case of an appropriately fine grid for an LES, as used by Sternel et al., the interface would be placed between $z^+ \approx$ 30–50. As the characteristics of the flow in the present combustor model deviate strongly from those encountered in an equilibrium flow, as e.g. in a fully-developed channel flow considered in Section 3, the latter values would indicate an interface being positioned too close to the wall, as it can be concluded from Fig. 7. Here, the evolution of the interface location separating the RANS and LES sub-regions in the framework of the present hybrid LES/RANS, performed at the coarsened grid, is displayed. The interface position expressed in terms of the non-dimensional wall distance was approximately $z^+_{ifce} \approx 200$. The average z^+ value along the combustor symmetry plane is about 1,800. However, there are regions, especially within the free recirculation zone, where z^+ values exceed 4,000. Besides using the present HLR model, the computations, applying the same grid, were also performed by applying the conventional LES (with the SGS model due to Smagorinsky) and the k–ε–ζ–f model. The present LES aims solely at comparing directly the predictive capabilities of both schemes. The computational time needed for the hybrid model is approximately 30% longer then for the coarse LES, due to solving two additional transport equations (for k and ε) and the interface treatment.

The flow topology in the present combustor configuration is illustrated by the time-averaged streamlines and the iso-contours of the axial velocity component in the lower half of the horizontal x–y plane, Fig. 8a. A strong influence of the swirling jets resulting in a closed recirculation zone in the core flow can be observed. This swirl-induced back-flow region penetrates into the combustor interior up to the position corresponding to $x/L \approx 0.0$. Its computationally (HLR) obtained length is about 130 mm. The strong swirl promotes the back-flow already within the air-blast nozzle (see also the experimental outcome in Fig. 6a for direct comparison). The free recirculation zone covers the largest part of the nozzle's central opening. Thus, a high velocity jet (up to 50 m/s) in the outer radial part of the nozzle outflow is generated. Lower boundary in Fig. 8a at $y/H = -0.5$ represents the side boundary plane of the solution domain being positioned in the middle between the presently accounted nozzle and the neighbouring nozzle (taken indirectly into account through the periodic boundary conditions). Between two nozzles a wall-bounded low velocity zone arises. The flow between two separation zones aligned with the mean dividing streamline, whose high velocity (up to 20 m/s) originates from the cross-section constriction, spreads gradually until the side boundaries are reached. In order to experimentally investigate the interaction between the swirling streams discharging

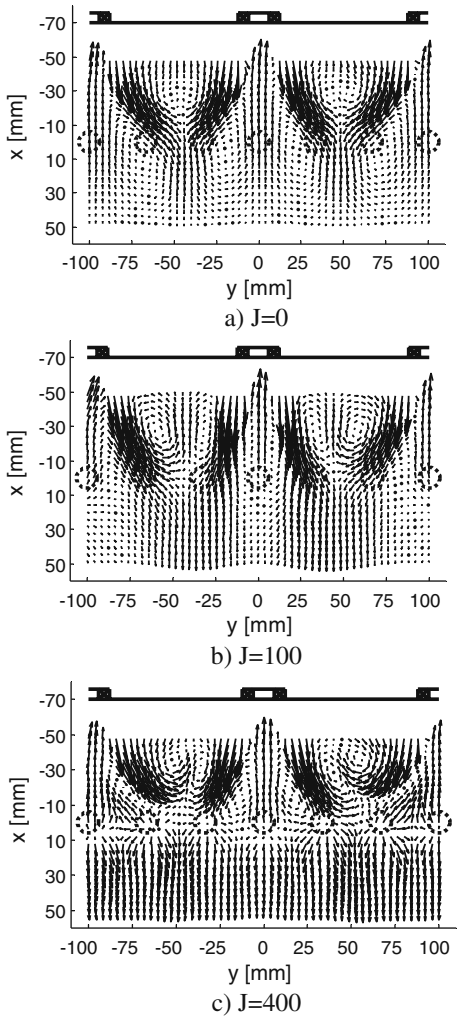

Fig. 6 Velocity vectors obtained by PIV measurements for several momentum flux ratios J in the horizontal combustor plane ($z = 0$)

the nozzle, the mixing chambers with three and five air-blast nozzles were built. It would have been desirable to consider more than one nozzle in the simulations, however, one can justify the application of the periodic boundary conditions between the nozzles when analysing the resulting velocity and stress profiles, Figs. 8b and 9. They display the comparison of the experimentally obtained axial velocity profiles, streamwise and shear stress components and corresponding computational results at six streamwise locations in the horizontal x–y plane (denoted in Fig. 1b). The velocity profile evolution exhibiting regions with negative velocity values in the mixing chamber core and positive peak values coinciding with the shear layer regions is a typical representative of the flow in a swirl combustor. The development of the Reynolds stress profiles within the combustor, whose maxima correspond to the regions with the most intensive mean flow deformation, are displayed in Fig. 9. The results, especially those obtained by the present HLR model follow closely the experimental

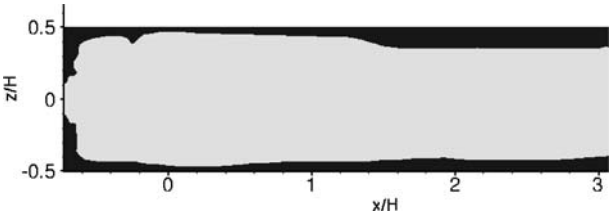

Fig. 7 Evolution of the interface (in the vertical plane $y = 0$) within combustor interior separating the RANS sub-region (*black area*) and the LES sub-regions (*grey area*) in the framework of the present hybrid LES/RANS. The interface position corresponds approximately to $z^{+}_{ifce} \approx 200$

ones. Proper capturing of the velocity profiles with respect to their local maxima and minima indicates the correctly predicted rate of spreading of the swirling jets and the form, size and intensity of the flow reversal zone. It is also in accordance with the rate of entrainment of the curved, annular swirling stream into the core flow. In contrast, the k–ε–ζ–f model returns velocity profiles indicating a continuous overprediction of flow rate in this central combustor region and magnification of the flow reversal in the region between current nozzle and the neighbouring ones accounted for by imposing the periodic conditions at the side planes of solution domain. The LES results

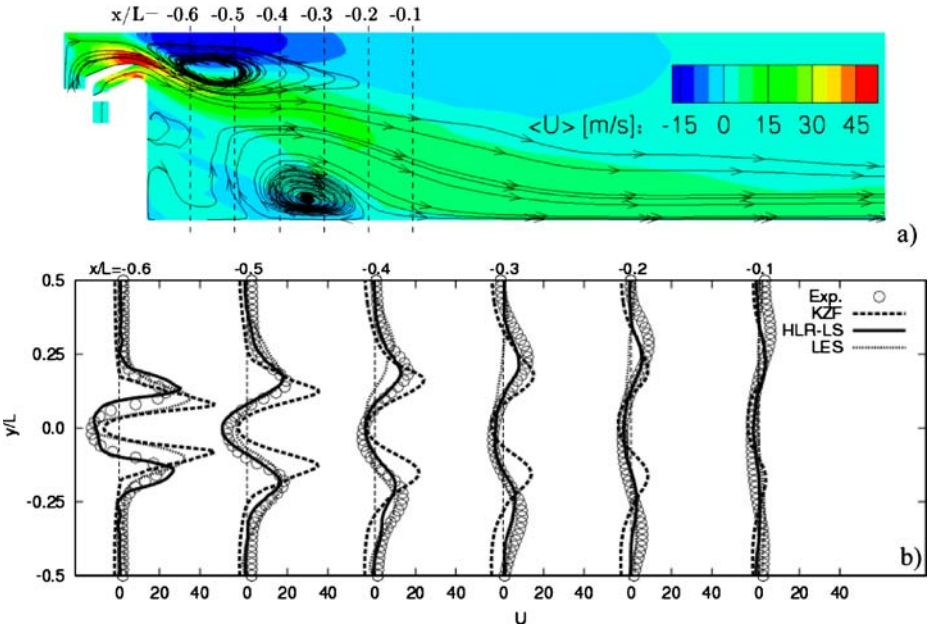

Fig. 8 Computationally (HLR) obtained time-averaged streamlines and axial velocity contours (**a**) and their profiles (**b**) at selected streamwise positions in the horizontal plane ($z = 0$) of the mixing chamber for the case without secondary air injection ($J = 0$). L is the solution domain width ($= w/3$; see Table 1). *KZF* denotes the k–ε–ζ–f RANS model; *HLR-LS* denotes the hybrid LES/RANS method employing the Launder–Sharma (LS) RANS model

exhibit significant deviation already at the location $x/L \approx -0.6$ situated immediately after the swirling flow expansion. This is manifested through a narrower central back-flow zone characterized by the velocity magnitude overprediction in the mixing layer region. Corresponding turbulence intensity, represented by the streamwise and shear Reynolds stress components (Fig. 9), confirms excessive turbulence production being compatible with steep mean velocity gradients. A plausible reason for such performance can be attributed to the incorrect capturing of the flow within the swirl nozzle, having its cause in the inadequate resolution of the wall boundary layer within the swirl generator section. This was very important keeping in mind that these wall layers unify in the combustor passing into the region of shear layer. After that, the LES velocity profile development exhibits strong decay of the velocity maximum indicating more intensive spreading of the central stream in contrast to the experimental observation. The outcome of the LES in conjunction with the coarse grid applied was not unexpected. The turbulent intensity overprediction is a typical indication for an inadequate grid resolution in the LES framework. This relates especially to the wall region within the combustor channel. The boundary layer at the combustor walls is subjected to the pressure gradient variation caused by alternating flow acceleration and deceleration influencing to a large extent the core flow. Impression about that could be gained by analysing Fig. 7. A narrower

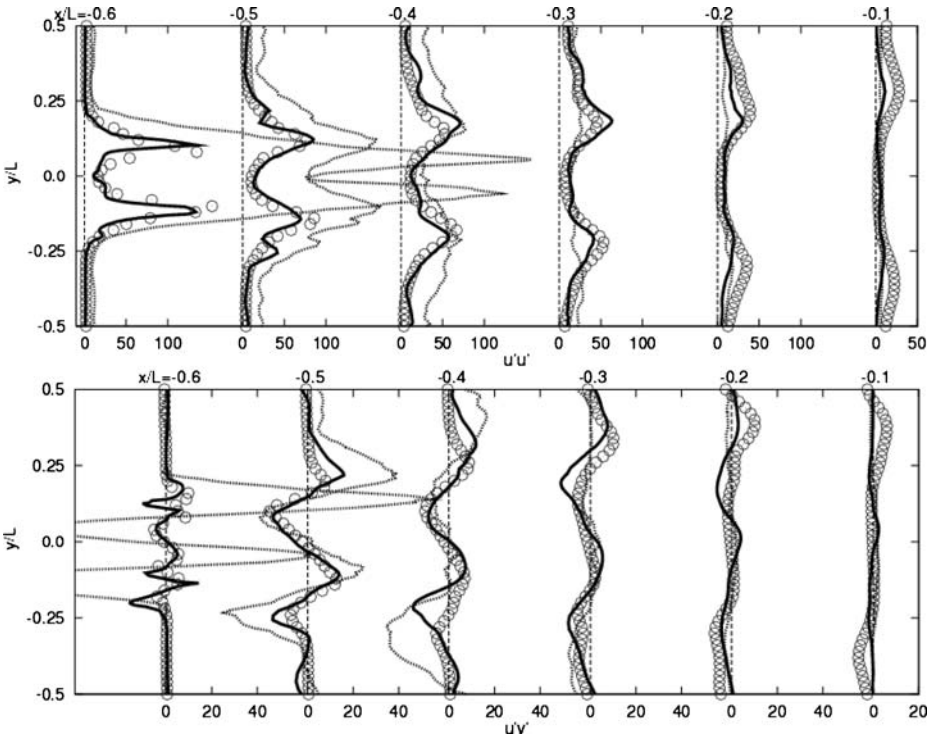

Fig. 9 Evolution of the streamwise and shear stress component profiles in the horizontal plane ($z = 0$) of the mixing chamber for the case without secondary air injection ($J = 0$). Legend as in Fig. 8

RANS sub-region corresponds to the boundary layer acceleration occurring in the region of the largest constriction of the positive axial flow due to presence of the free recirculation zone. After that a certain flow relaxation follows, being manifested by its deceleration towards the right end of the combustor channel (thicker RANS sub-region). Capturing all these phenomena by employing conventional LES requires appropriate grid resolution also in the streamwise and spanwise directions. A zonal hybrid LES/RANS method is less sensitive, as illustrated in Figs. 8 and 9. Despite the fairly coarse grid being used, the results obtained by the HLR method exhibit reasonable agreement with measurements in all characteristic flow regions, in contrast to the LES results obtained with the same grid.

A general impression about the vortex structure in the combustor at the swirl generator exit can be obtained by analyzing the Q-parameter ($Q = 0.5(\Omega_{ij}\Omega_{ij} - S_{ij}S_{ij})$) illustrated in Fig. 10a. Here, the Q-value corresponding to -5×10^4 s^{-2} is presented. A helical structure surrounding the recirculation zone is associated with the revolving motion of the central vortex around the symmetry axis. It indicates the well-known Precessing-Vortex-Core (PVC) representing a periodical, axially asymmetrical instability, typical for swirl intensities above approx. 0.6. Figure 10b displays the one-dimensional spectra of the velocity fluctuations illustrating their energy content over two frequency decades. In the lower frequency range the energy level captured computationally agrees reasonably with the experimental findings. The afore-mentioned PVC phenomenon is characterized by a certain (dominant) frequency corresponding to the peak at the power spectral density function for the

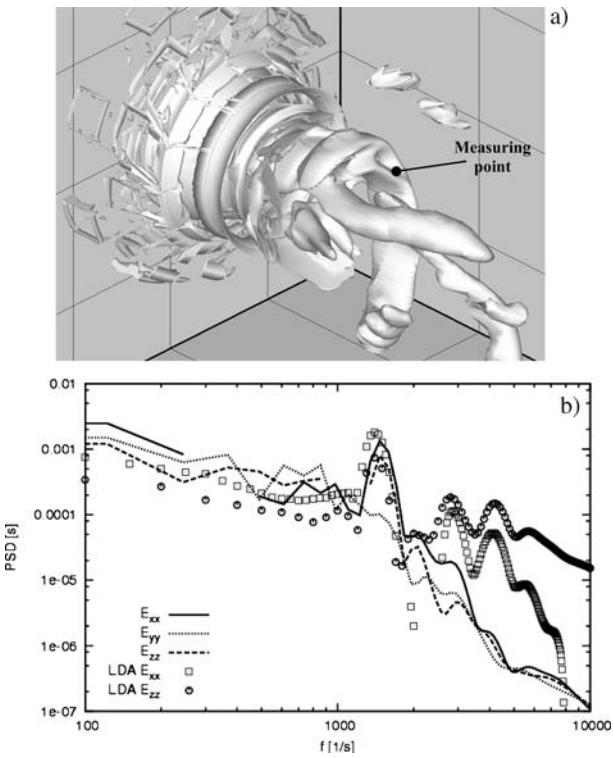

Fig. 10 Computationally (HLR-LE) obtained vortex structure visualized by the Q-criterion (**a**) and spectral density functions for velocities (**b**) recorded at the denoted point for the case without secondary air injection ($J = 0$). Consult Fig. 1 for coordinate directions

streamwise (x-direction) and normal (z-direction) velocity components, measured at the point situated in the shear layer (in the horizontal x–y plane; no peak pertinent to the velocity component in y-direction could be measured at this point). The computationally obtained frequency corresponds to $f \approx 1{,}470$ Hz being very close to the experimentally determined one $f_{PVC} \approx 1{,}450$ Hz (Fig. 3). Furthermore, the entire frequency range characterizing the PVC phenomenon (between 1,100 Hz and 2,000 Hz) as well as the corresponding energy content exhibit good agreement with the measured values.

Combustor with secondary air injection ($J = 100$) For the computations of this second configuration three holes on both the lower and upper walls of the mixing chamber are added, through which the secondary air is introduced into the primary zone. Accordingly, a new computational mesh requiring a certain refinement in the regions of secondary jets had to be appropriately designed. Similar to the first case (with $J = 0$) the original fine-LES grid, consisting of approximately 7 million grid cells (used for the simulations of the case with $\dot{m} = 50\ \mathrm{g/s}$), was suitably coarsened with respect to its use within the hybrid LES/RANS framework (see related discussion in the part concerning the first case). This coarse grid comprises 870,000 grid cells in total. The secondary air injection was considered by prescribing the air jet velocity direct at the hole opening. Its value $\overline{\mathrm{v}}_{SF} = 16.6\ \mathrm{m/s}$ is obtained in accordance with the prescribed momentum flux ratio ($J = 100$) and the mass flow rate of the main flow $(\dot{m} = 20\ \mathrm{g/s})$, i.e. the corresponding bulk velocity $\overline{\mathrm{v}}_{MF} = 1.66\ \mathrm{m/s}$. The experimentally observed contraction of the effective jet cross-section within the secondary air hole (from 12 mm to 10 mm) was accounted for in the computational framework. The other boundary conditions correspond to those applied in the case with $J = 0$. It turned out in the course of the computational campaign that the wall-closest computational nodes was situated between $z^+ \approx 15$–25. It is well known that the exact wall boundary conditions can be applied only if the wall-adjacent node is situated within the viscous sublayer characterized by the linear velocity profile ($U^+ = z^+$). In the case of the flow subjected to sudden expansion the edge of this viscosity-influenced region is at $z^+ \approx 2$–3. Accordingly the near-wall, eddy-viscosity-based RANS model couldn't be applied in the RANS sub-region, but its high-Reynolds number counterpart. Consequently, the standard wall functions were used for near-wall treatment. On the other hand, one knows that the wall functions are derived under the assumptions of an equilibrium flow, underlying to the logarithmic law for the velocity field, which is not encountered

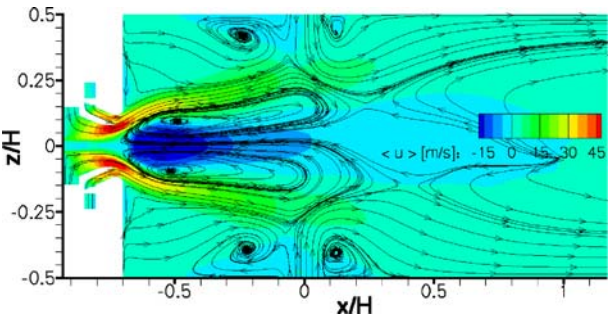

Fig. 11 Computationally (HLR-KE) obtained time-averaged streamlines and axial velocity contours in the vertical mixing chamber plane $y = 0$. H is the main channel height ($= h_{MF}$; see Table 1). *HLR-KE* denotes the hybrid LES/RANS method employing the $k - \varepsilon$ RANS model

here. Underprediction of the velocity values corresponding to the logarithmic law is a typical departure for a velocity field subjected to the adverse pressure gradient effects due to sudden expansion, e.g. Hanjalić and Jakirlić [11]. However, the actual velocity profile follows also the logarithmic law but with a somewhat reduced value of the log-law coefficient. It justifies to a certain extent the employment of the standard wall functions for modelling the near-wall region in the configuration considered. The HLR interface is located at $z/\Delta \approx 2$. In the case of a uniform grid it would result in only two grid cells within the RANS layer. However, the non-uniform grid squeezed gradually by approaching the walls applied in the present simulation implies about six to seven grid cells being accommodated within the RANS sub-region of the flow domain. The pure RANS computations using the same k–ε model (KE) were also performed.

An impression about the flow topology can be obtained by analysing the computationally (HLR-KE) obtained time-averaged streamlines and iso-contours of the axial velocity component in the vertical plane ($y = 0$) of the mixing chamber, containing also the secondary hole outlets, Fig. 11. Similar to the previous case, the free back-flow region will be created following the swirled outflow from the air-blast nozzle. The bulk velocity of the ring-shaped shear layer surrounding the

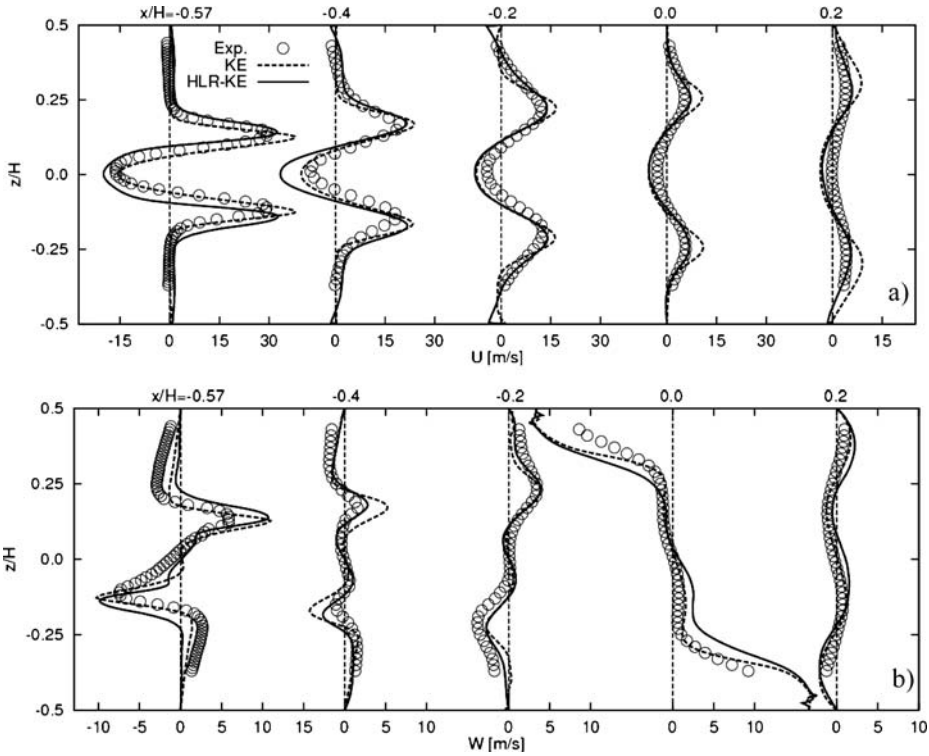

Fig. 12 Computationally obtained profiles of the axial (**a**) and normal (**b**) mean velocity components at selected streamwise positions in the vertical mixing chamber plane $y = 0$. *KE* denotes the standard $k - \varepsilon$ RANS model; *HLR-KE* denotes the hybrid LES/RANS method employing the $k - \varepsilon$ RANS model

separation bubble of approx. 16 m/s corresponds exactly to the secondary air jet velocity $\overline{v}_{SF}$. These two streams are equally deflected after impinging onto each other. Thereby, the spreading of the inner recirculation zone towards combustor walls, i.e. the evolution of its size, is suppressed to a certain extent. The same outcome is obtained experimentally (see Fig. 12). Whereas the secondary air jets are skewed towards the combustor exit and continue parallel to its walls, the major part of the ring-shaped main stream flows back into the recirculation zone. The local separation regions around the secondary jets are also recognizable in Figs. 11 and 12a.

Figures 12 and 13 display the experimentally and computationally obtained evolution of the profiles of the axial and normal (z-direction) velocities, streamwise and shear stress components in the vertical mixing chamber plane ($y = 0$). Characteristic streamwise locations ahead of and after the injection plane were selected. One should note that the experimental data are not available in the immediate wall vicinity ($z/H \geq 0.45$). The profile shapes of all quantities characterized by local minima and maxima and sign changes (the latter is pertinent to the velocity components) are similar to those documented in the previous configuration. The suppression of the free recirculation zone illustrated in the computationally obtained streamline pattern (Fig. 11) is reflected in the intensification of the flow reversal (corresponding to a larger negative velocity peak in Fig. 12a) and the flow in the shear layer region.

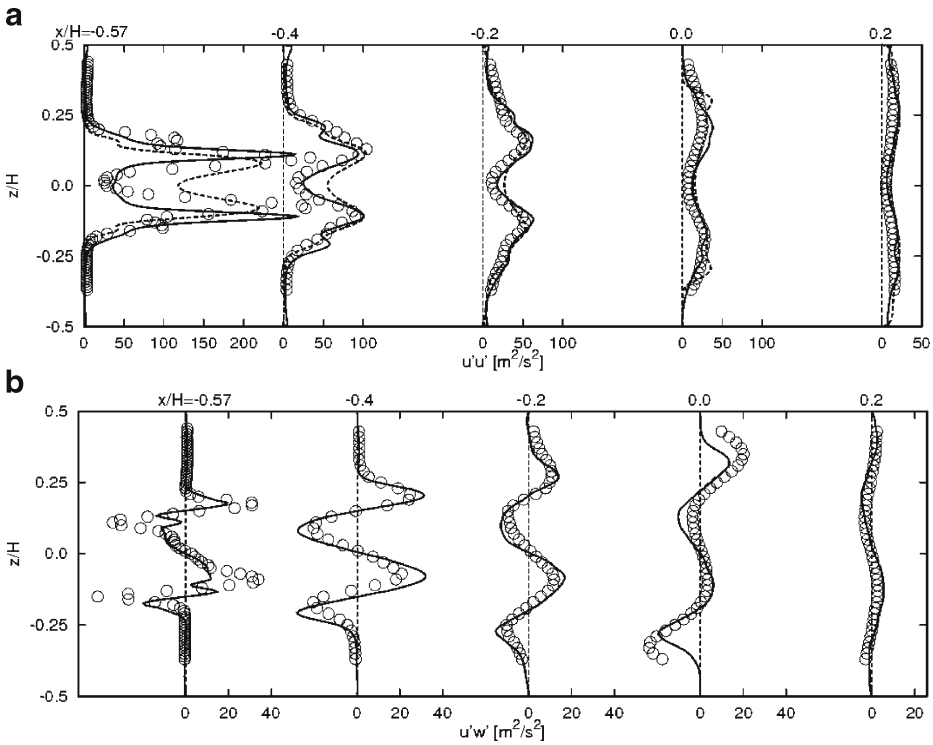

Fig. 13 Evolution of the profiles of the streamwise (**a**) and shear (**b**) stress components in the vertical mixing chamber plane ($y = 0$) for the case with secondary air injection ($J = 100$). Legend as in Fig. 12

Such an enhanced mean flow deformation contributes strongly to the turbulence production resulting in large peaks of the turbulence stress components (Fig. 13), the situation being compatible with the previously investigated flow configuration. The time-averaged structure of the flow field is well reproduced by both computational methods applied. Apart from somewhat higher negative axial velocity values and a somewhat weaker intensity of the suppression of the central back-flow region characterized by an overprediction of the vertical velocity component in the shear layer (both departures being pertinent to the near field), the quantitative agreement with available experimental results is very good, especially with respect to the HLR method. The same is valid with respect to the turbulence field. An "unexpectedly" good agreement obtained by applying the standard k–ε model can be explained by the fact, that the large expansion ratio (the ratio of the mean channel height to the outlet nozzle diameter is about 4), associated with the strong adverse pressure

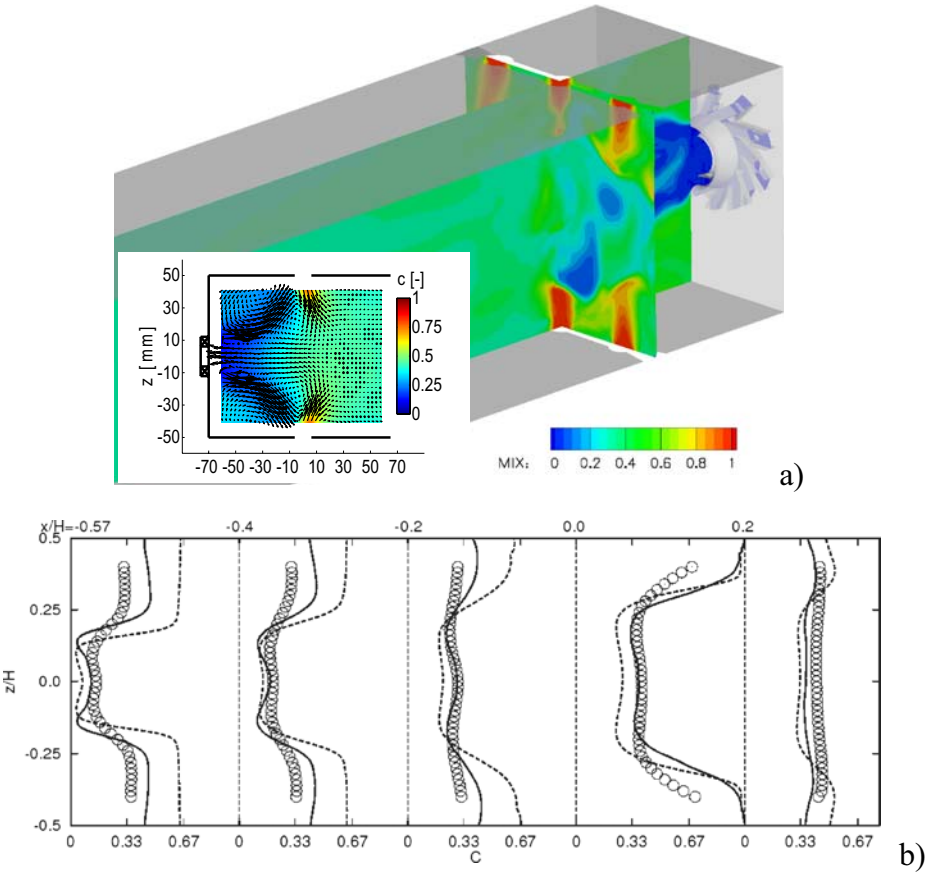

Fig. 14 Computationally (HLR) obtained time-averaged iso-contours of the mixture fraction (**a**) for the case with secondary air injection, $J = 100$ (figure in the middle shows the experimentally obtained velocity and concentration results) and evolution of the concentration profiles (**b**) in the vertical mixing chamber plane ($y = 0$) for the case with secondary air injection ($J = 100$). Legend as in Fig. 12

gradient effects (dominating over the stress gradients in the equation of motion relativizing their influence), acts towards Reynolds stress anisotropy weakening, i.e. towards promotion (to a certain extent) of an isotropic state of the turbulence (the eddy-viscosity models are based upon), benefiting in such a way its good predictive performance. There is a detailed evidence in the published literature about the latter phenomena, e.g. Wennerberg and Obi [39] and Jakirlić et al. [15]. The results of the concentration field are shown in Fig. 14. The prescribed inflow values of the concentration field $C = 0.0$ and $C = 1.0$ correspond to the swirl nozzle interior and secondary hole outlets ($x/H = 0.0$) respectively. Based on the contour plots analysis, it is clearly evident that the regions of the high shear strain and high gradient of the local concentration coincide and that the mixing between the main flow and secondary air jets is completed at the location $x/H = 0.2$. The corresponding value of the mixture fraction C is about 0.4. In the combustor near field ($x/H < 0.0$) values about 0.35 prevail, being reasonably well reproduced by the present HLR model (it is especially the case with respect to the core flow, $-0.25 \leq z/H \leq +0.25$), despite somewhat steeper gradients in the region of the ring-shaped main flow stream ($z/H \approx 0.25$).

5 Conclusions

The experimental and computational study of the flow in the mixing chamber of a single annular combustor model has been performed. Two flow configurations, without and with secondary air jet injection into the primary combustor zone, were investigated. The computational part focussed on the performance of a zonal hybrid LES/RANS (HLR) method, merging an eddy-viscosity-based RANS model resolving the near-wall region with the LES covering the outer layer (flow core). The coupling between the two sub-regions is realized by interchanging velocities and modelled turbulent quantities: kinetic energy of turbulence, its dissipation rate and turbulence viscosity. Large attention was paid to two key questions concerning the matching of both methods: the exchange of the variables across the LES/RANS interface (being realized by implicit imposition of the condition of equality of the modelled turbulent viscosities) and the interface position control (depending on the portion of the SGS turbulence kinetic energy to the total one). Hereby, great importance is attached to simplicity, efficiency and applicability to complex geometries.

Concerning the result presentation the emphasis was placed on the evolution of the time-averaged velocity and turbulence fields reflecting the structure of the swirl-induced recirculation zone and its interaction with the penetrating air jets. The development of the secondary air concentration under the described conditions was also the topic of the result analysis. The comparison with the experimental database displays good agreement of the results obtained by the present hybrid model scheme in all characteristic flow regions in the mixing chamber considered. This is especially the case with respect to a suppression of the opening angle of the ring-shaped shear layer bordering the recirculation zone due to secondary jets injection, whose effect is manifested in an appropriate deformation of the velocity field corresponding to the intensified back-flow. In addition to the correctly captured spatial evolution of the velocity, turbulence and concentration fields, comparative analysis of the time-

resolved images with respect to the precessing vortex core (PVC) phenomenon associated with the swirled main flow has been performed. It reveals good agreement between measurements and computations, especially pertaining to its high-frequency oscillation of $f_{PVC} \approx 1,450 - 1,470\,\text{Hz}$.

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References

1. Abe, H., Kawamura, H., Matsuo, Y.: Surface heat-flux fluctuations in a turbulent channel flow up to $Re_\tau = 1020$ with $Pr = 0.025$ and 0.71. Int. J. Heat Fluid Flow **25**, 404–419 (2004)
2. Albrecht, H.-E., Borys, M., Damaschke, N., Tropea, C.: Laser Doppler and Phase Doppler Measurement Techniques. Springer, Berlin (2002)
3. Böhm, B., Findeisen, J., Dreizler, A., Hennecke, D.K.: Untersuchung der Turbulenzstruktur einer verdrallten Brennkammerströmung, GALA, Karlsruhe, Germany, 7–9 September 2004
4. Böhm, B., Dreizler, A., Gnirß, M., Tropea, C., Findeisen, J., Schiffer, H.-P.: Experimental investigation of turbulence structure in a three-nozzle combustor. Paper No. GT2007-27111 (2007)
5. FASTEST-Manual: Chair of Numerical Methods in Mechanical Engineering. Department of Mechanical Engineering, Technische Universität Darmstadt, Germany (2005)
6. Findeisen, J., Gnirß, M., Damaschke, N., Schiffer, H.-P., Tropea, C.: 2D-concentration measurements based on Mie scattering using a commercial PIV system. In: Proc. 6th Int. Symposium on Particle Image Velocimetry, Pasadena, USA, 21–23 September 2005
7. Gan, X.P.: Experimental and analytical studies of jets in quiescent or rotating flow fields. Ph.D. thesis, University of Bath, UK (1990)
8. Gnirß, M., Findeisen, J., Damaschke, N., Tropea, C., Schiffer, H.-P.: Experimental investigation of coherent flow structures in a gas-turbine combustor model. In: Proc. 5th Int. Symp. on Turbulence, Heat and Mass Transfer, Dubrovnik, Croatia, 25–29 September 2006
9. Gnirß, M., Tropea, C.: Simultaneous PIV and concentration measurements in a gas-turbine combustor model. Exp. Fluids **45**(4), 643–656 (2008)
10. Gnirß, M.: Strömung und Mischung im Primärzonenbereich von modernen Gasturbinenbrennkammern. Ph.D. thesis, Technische Universität Darmstadt, Germany, Shaker Verlag, ISBN 978-3-8322-7821-2 (2008)
11. Hanjalić, K., Jakirlić, S.: Contribution towards the second-moment closure modelling of separating turbulent flows. Comput. Fluids **27**(2), 137–156 (1998)
12. Hanjalić, K.: Hybrid RANS-LES—a Critical Appraisal and a Proposal. Internal Report, Department of Aeronautics, Imperial College London, UK and Technical University Delft, The Netherlands (2001)
13. Heitor, M.V., Whitelaw, J.H.: Velocity, temperature and species characteristic of the flow in a gas-turbine combustor. Combust. Flame **64**, 1–32 (1986)
14. Hanjalić, K., Popovac, M., Hadziabdić, M.: A robust near-wall elliptic-relaxation eddy-viscosity model for CFD. Int. J. Heat Fluid Flow **25**, 1047–1051 (2004)
15. Jakirlić, S., Hanjalić, K., Tropea, C.: Modelling rotating and swirling flows; a perpetual challenge. AIAA J. **40**(10), 1984–1996 (2002)
16. Jakirlić, S., Kniesner, B., Šarić, S., Hanjalić, K.: Merging near-wall RANS models with LES for separating and reattaching flows. In: ASME Joint U.S.–European Fluids Engineering Summer Meeting: Symposium on "DNS, LES and Hybrid RANS/LES Techniques", Miami, FL, USA, 17–20 July, Paper No. FEDSM2006-98039 (2006)
17. Jester-Zürker, R., Jakirlić, S., Tropea, C.: Computational modelling of turbulent mixing in confined swirling environment under constant and variable density conditions. Flow Turbul. Combust. **75**(1–4), 217–244 (2005)
18. Kniesner, B., Jester-Zürker, R., Jakirlić, S., Hanjalić, K.: RANS-SMC and hybrid LES/RANS of a backward-facing step flow subjected to increasingly enhanced wall heating. In: 5th International

Symposium on Turbulence and Shear Flow Phenomena (TSFP5), Munich, Germany, 27–29 August 2007
19. Kniesner, B.: Ein hybrides LES/RANS Verfahren für konjugierte Strömung, Wärme- und Stoffübertragung mit Relevanz zu Drallbrennerkonfigurationen (A hybrid LES/RANS method for conjugated flow, heat and mass transfer with relevance to swirl combustor configurations). Ph.D. thesis, Technische Universität Darmstadt, Germany (2008). http://tuprints.ulb.tu-darmstadt.de/950/
20. Koutmos, P.: An isothermal study of gas turbine flows. Ph.D. thesis, University of London (1985)
21. Koutmos, P., McGuirk, J.J.: Isothermal flow in a gas-turbine combustor—a benchmark experimental study. Exp. Fluids **7**(5), 344–354 (1989)
22. Koutmos, P., McGuirk, J.J.: Isothermal modeling of gas-turbine combustors—computational study. J. Propuls. Power **7**(6), 1064–1071 (1991)
23. Launder, B.E., Sharma, B.I.: Application of the energy-dissipation model of turbulence to the calculation of flow near a spinning disc. Lett. Heat Mass Transf. **1**, 131–138 (1974)
24. Lin, C.A.: Three-dimensional computation of injection into swirling cross flow using second-moment closure. Ph.D. thesis, Faculty of Technology, University of Manchester (1990)
25. Lin, C.A., Leschziner, M.A.: Three-dimensional computation of transient interaction between radially injected jet and swirling cross-flow using second-moment closure. Comput. Fluid Dyn. J. **1**(4), 419–428 (1993)
26. Mason, P.J., Callen, N.S.: On the magnitude of the subgrid-scale eddy coefficient in large-eddy simulation of turbulent channel flow. J. Fluid Mech. **162**, 439–462 (1986)
27. McGuirk, J.J., Palma, J.M.L.M.: Experimental investigation of the flow inside a water model of a gas turbine combustor: part 1—mean and turbulent flowfield. ASME J. Fluids Eng. **117**(3), 450–458 (2002)
28. Monmont, F., Greenhalgh, D.: Cranfield University Contribution to Final Report on Brite-Euram III Project LES4LPP. Technical Report LES4LPP-BE95-1953 (1999)
29. Mueller, C.M., Knill, K.J.: Numerical simulation of emissions in a gas turbine combustor. Combustion Canada '96 (1996)
30. Palm, R.: Experimentelle Untersuchung der Strömung und Vermischung in einem Drallbrennermodell (Experimental investigation of flow and mixing in a model of swirl combustor). Ph.D. thesis, Technische Universität Darmstadt, Germany (2006). http://tuprints.ulb.tu-darmstadt.de/723/
31. Palm, R., Grundmann, S., Weismüller, M., Šarić, S., Jakirlić, S., Tropea, C.: Experimental characterization and modelling of inflow conditions for a gas turbine swirl combustor. Int. J. Heat Fluid Flow **27**(5), 924–936 (2006)
32. Pierce, C.D., Moin, P.: Method for generating equilibrium swirling inflow conditions. AIAA J. **36**(7), 1325–1327 (1998)
33. Pope, S.: Turbulent flows. Cambridge University Press, ISBN 0-521-59886-9 (2000)
34. Roback, R., Johnson, B.V.: Mass and Momentum Turbulent Transport Experiments with Confined Swirling Coaxial Jets. NASA Contractor Report 168252 (1983)
35. Sternel, D.C., Schäfer, M., Gauß, F., Jigit, S.: Influence of numerical parameters for LES of complex flow fields. In: Wesseling, P., Oñate, E., Périaux, J. (eds.) European Conference on Computational Fluid Dynamics, ECCOMAS CFD 2006. TU Delft, The Netherlands (2006)
36. Tang, G., Yang, Z., McGuirk, J.J.: Large eddy simulation of isothermal confined swirling flow with recirculation. In: Rodi, W., Fueyo, N. (eds.) Proc. 5th Int. Symp. on Engineering Turbulence Modelling and Measurements, pp. 885–894. Elsevier Science Ltd. (2002)
37. Temmerman, L., Hadziabdic, M., Leschziner, M.A., Hanjalic, K.: A hybrid two-layer URANS-LES approach for large eddy simulation at high Reynolds numbers. Int. J. Heat Fluid Flow **26**, 173–190 (2005)
38. Voigt, P., Schodl, R., Griebel, P.: Using the laser light sheet technique in combustion research. In: 90th Symp. of AGARD–PEP on Advanced Non-intrusive Instrumentation for Propulsion Engines (1997)
39. Wennerberg, D., Obi, S.: Prediction of strongly swirling flows in quarl expansions with a non-orthogonal finite-volume method and a second-moment turbulence closure. In: Rodi, W., Martelli, F. (eds.) Engineering Turbulence Modelling and Experiments, vol. 2, pp. 197–206. ISBN 0444898026 (1993)